Network Layer:
Internet Protocol, Version 6 (IPv6)

Lecture 14 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

IPv6

- Initial motivation:
  - 32-bit address space soon to be completely allocated.
- Additional motivation:
  - Simpler header format helps speed processing/forwarding
  - header changes to facilitate QoS

Reasons for using IPv6

- Address Availability:
  - IPv4: 4 octets - 32 bits
    - 2^32 or 4,294,467,295 IP Addresses.
  - IPv6: 16 octets - 128 bits
    - 3.4 x 10^38 or

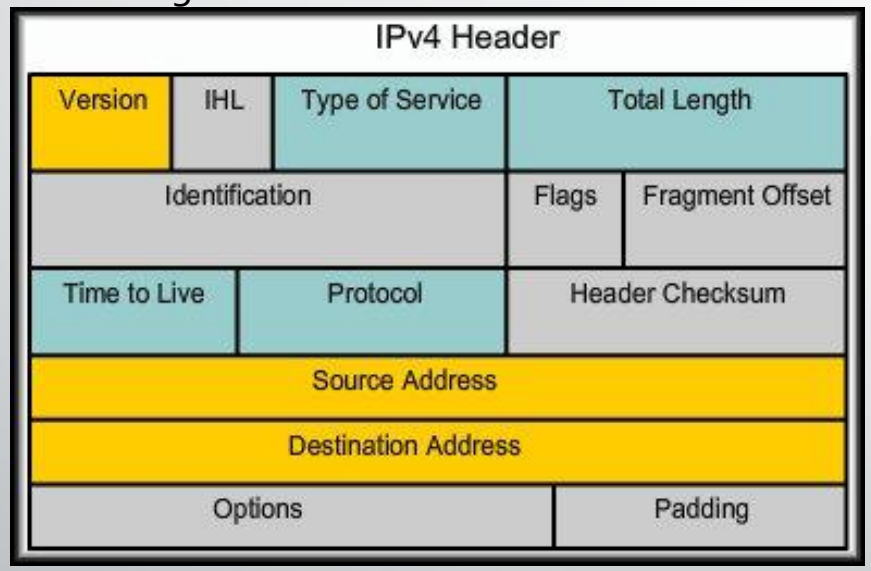
340,282,366,920,938,463,463,374,607,431,768,211,456

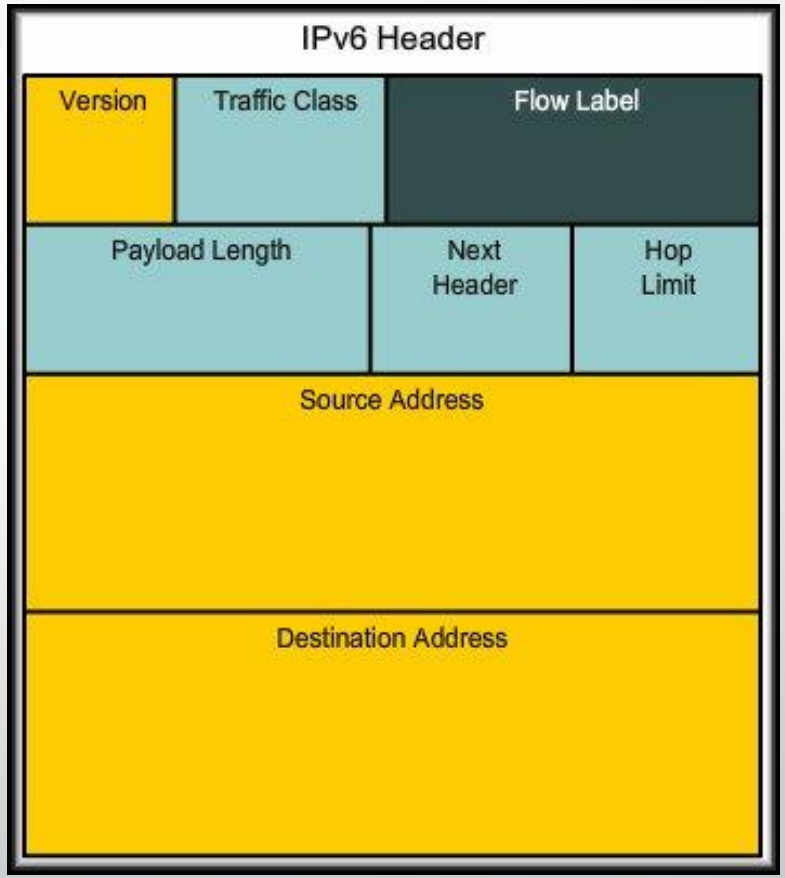
(340 undecillion) IP Addresses.

- *Every atom of every person on Earth could be assigned 7 unique addresses with some to spare (assuming*
  7×10^{27} atoms per human x 6.5 Billion).

Reasons for Using IPv6

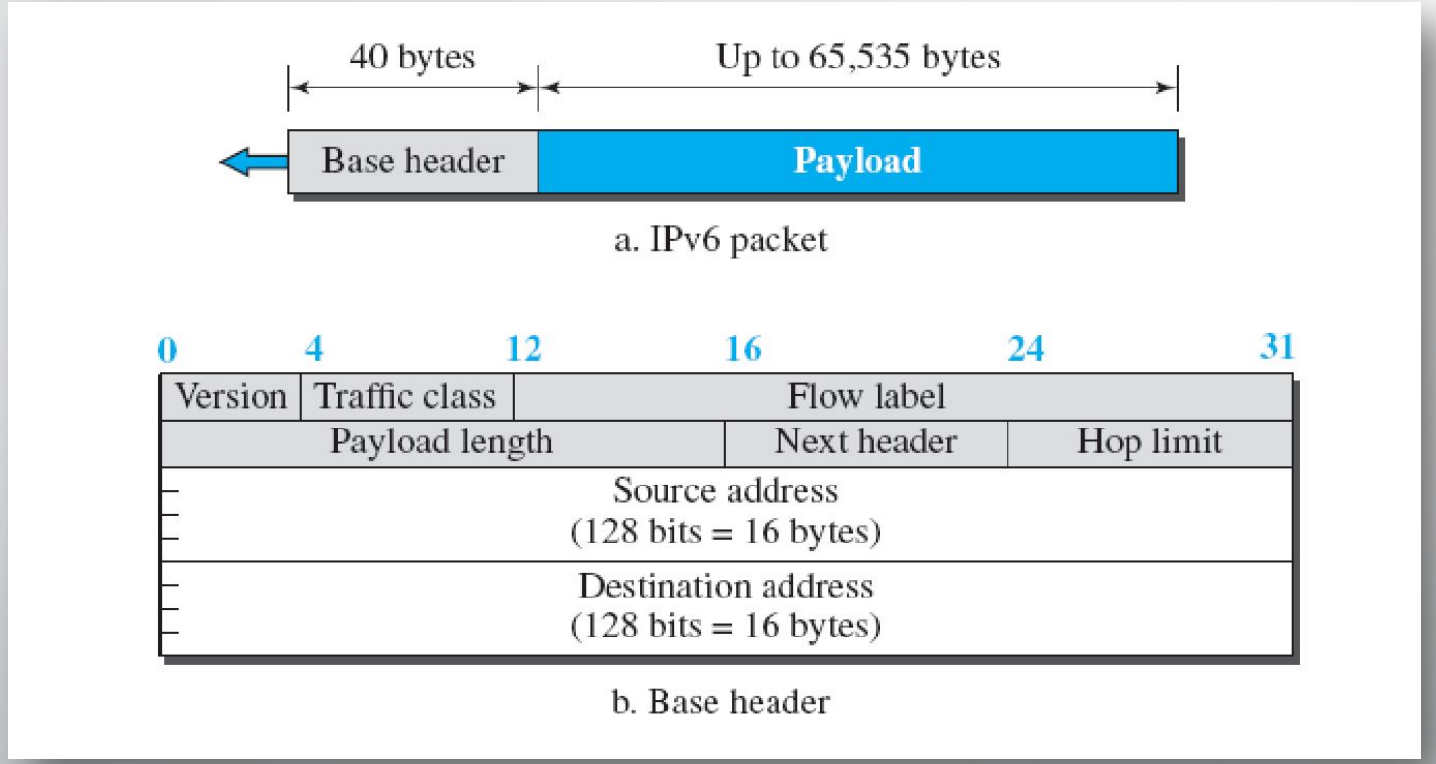
- IPv6 Features:
  - fixed-length 40 byte header
  - no fragmentation allowed



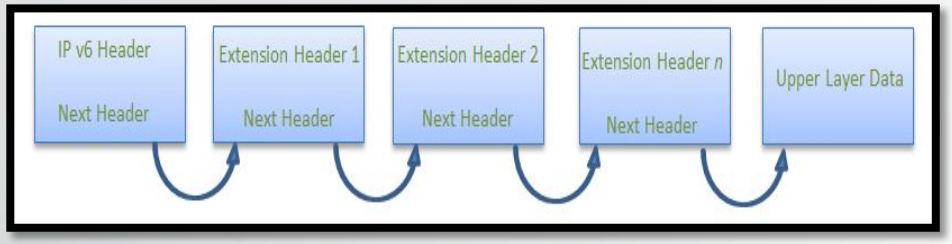


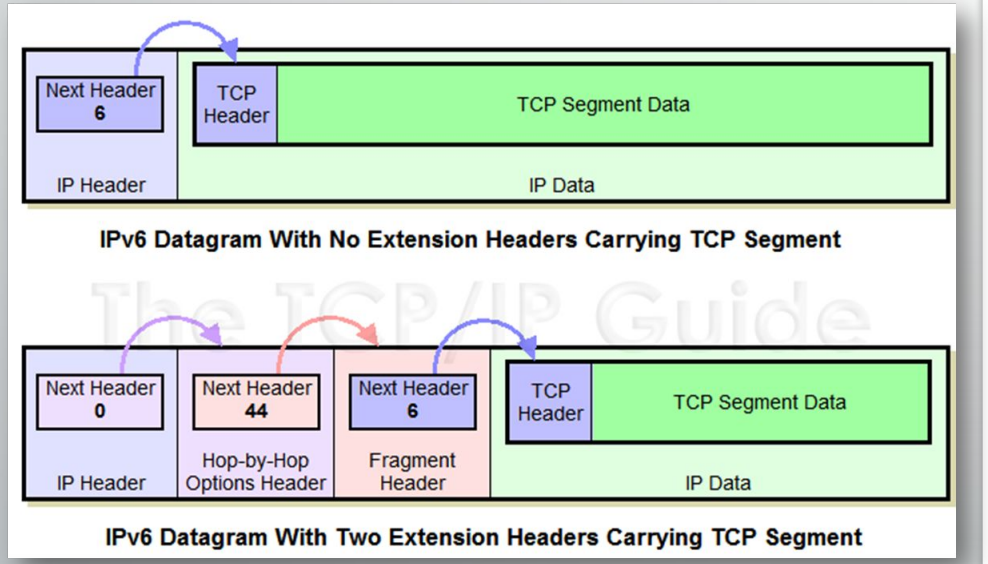
IPv6 Datagram

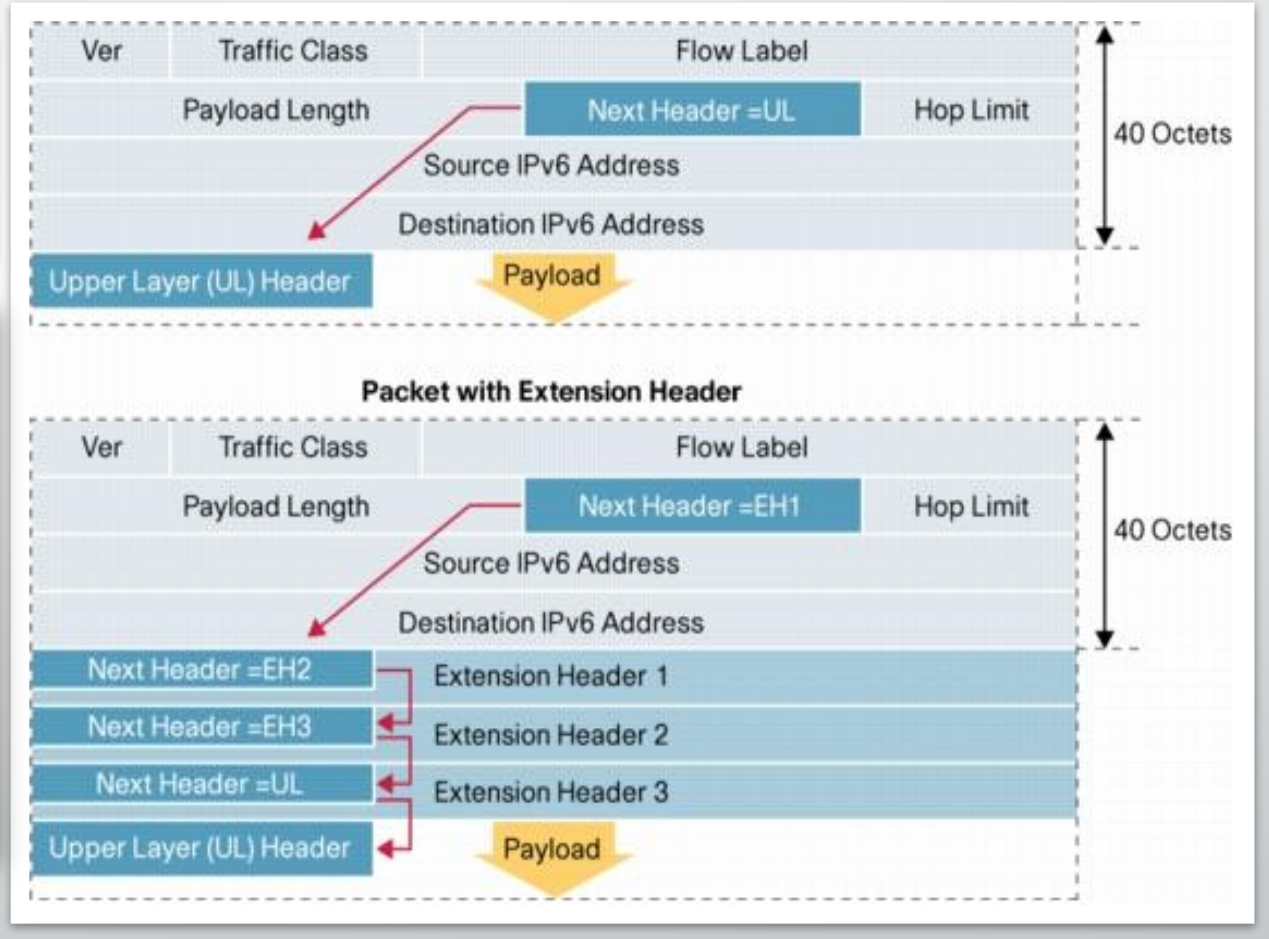
40 Octets, 8 fields



Extension Headers

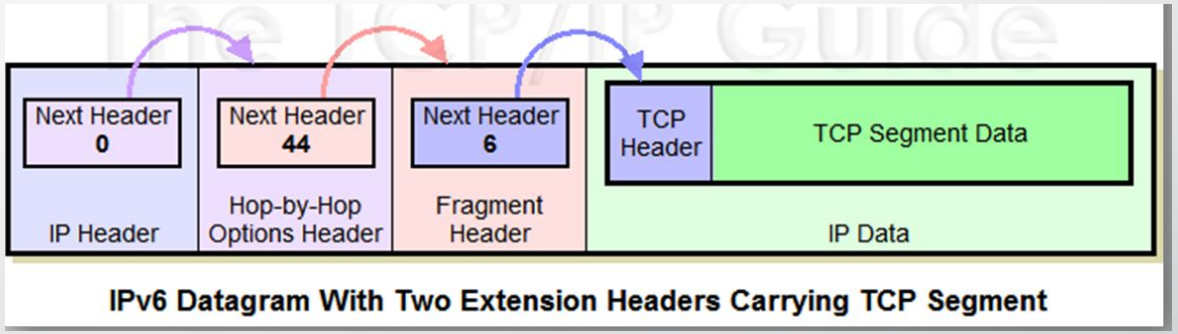






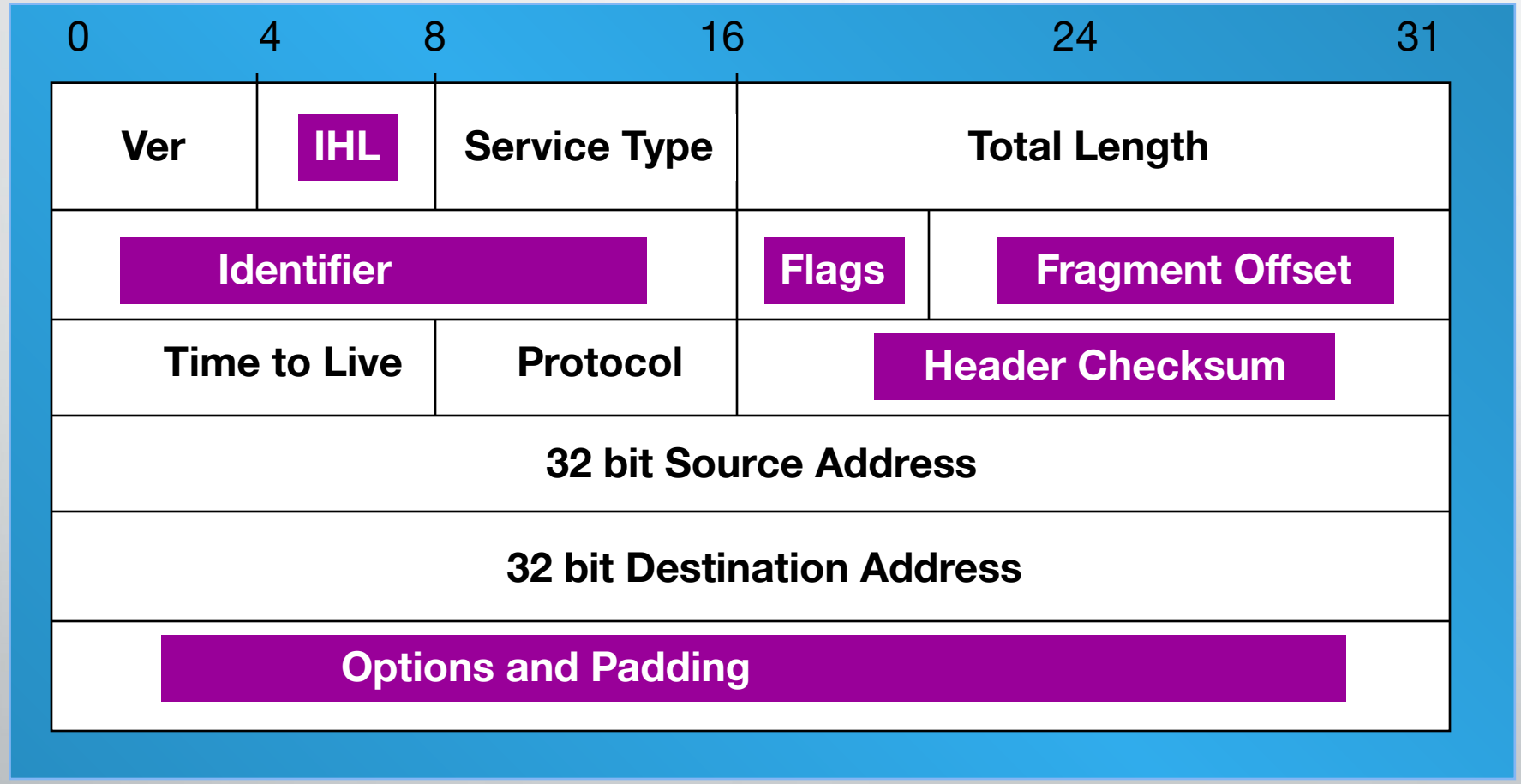
Extension Headers

- Basic header simplified for ease of processing
- Additional information carried in extension headers
  - Hop-by-hop options
  - Routing header
  - Fragment header
  - Destination options header
  - Authentication header (AH)
  - Encrypted security payload (ESP) header
- Next Header field says what type of header follows
  - E.g. Fragment Header, TCP, ICMP, etc.



IPv6 Datagram With Two Extension Headers Carrying TCP Segment

The IPv4 Header



shaded fields are absent from IPv6 header

Header Changes between IPv4 and IPv6

- Revised
  - Time to Live (Hop Limit)
  - Addresses increased from 32 bits to 128 bits
  - Protocol (Next Header)
  - Precedence & TOS (Traffic Class)
- Extended
  - Flow Label field added (Recommended read: Page 676 of Forouzan's Book)

IPv6 Address

- 128 bits
- given below is a 128 bit IPv6 address represented in binary format and

```
0010000000000001 0000000000000000 0011001000111000 1101111111100001
0000000001100011 0000000000000000 0000000000000000 1111111011111011
```

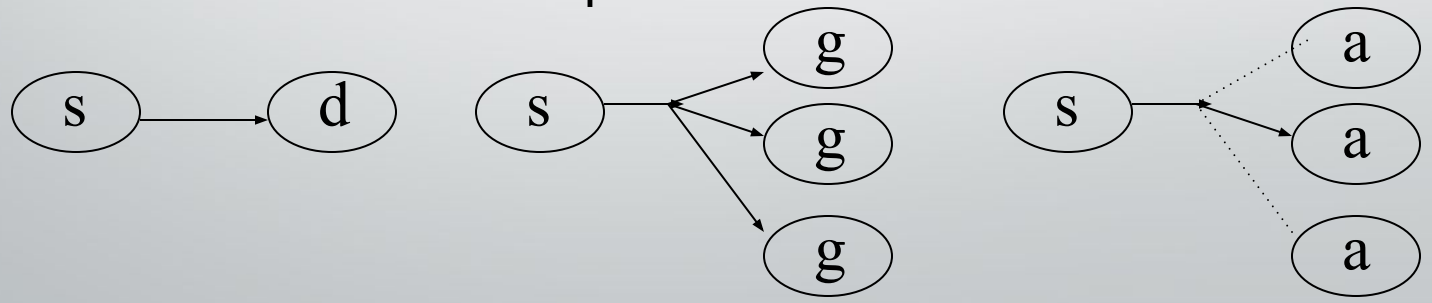
-

```
2001:0000:3238:DFE1:0063:0000:0000:FEFB
```

Called **string notation**

Types of IPv6 addresses

- **unicast**
  - communicate specified 1 computer
- **multicast**
  - communicate group of computers
- **anycast**
  - send group address that can receive multiple computers, but receive 1 computer



unicast multicast anycast

Types of IPv6 addresses

- Unlike IPv4, there is no broadcast address.
- There is an "all nodes multicast" which serves the same purpose.

Unicast Global Addresses

- These are assigned by the IANA and used on public networks.
- They are equivalent to IPv4 global (sometimes called public) addresses.
- Typically they start at 2000::/3

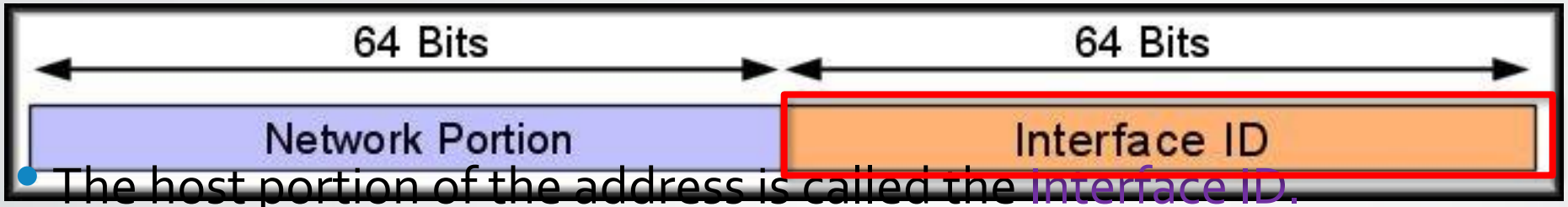
Unicast addresses

- A unicast address is an address that identifies a single device.

- Types of Unicast Addresses:

Block prefix	*CIDR*	*Block assignment*	*Fraction*
0000 0000	0000::/8	Special addresses	1/256
001	**2000::/3**	**Global unicast**	**1/8**
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256

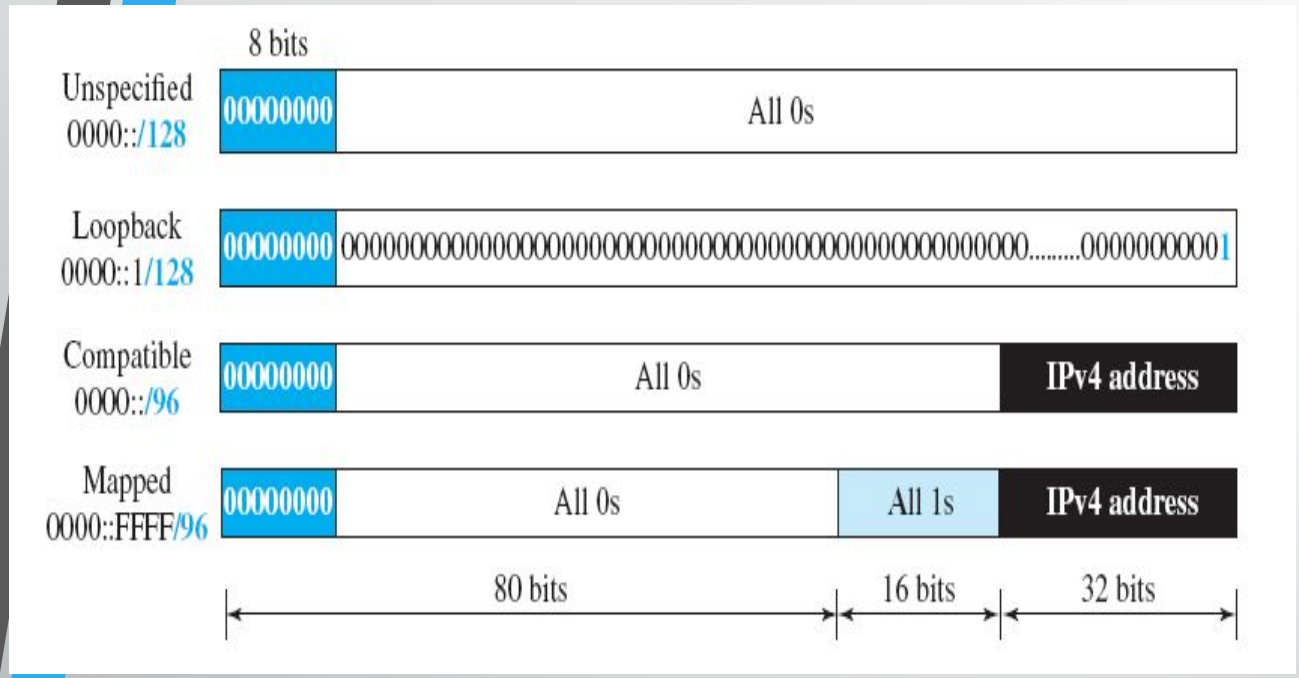
Unicast Global Addresses



- The host portion of the address is called the Interface ID.

- Can contain:
  - The interface's 48-bit MAC Address.
  - An identifier derived from the EUI-64 Address (more later).
  - A manually configured address.

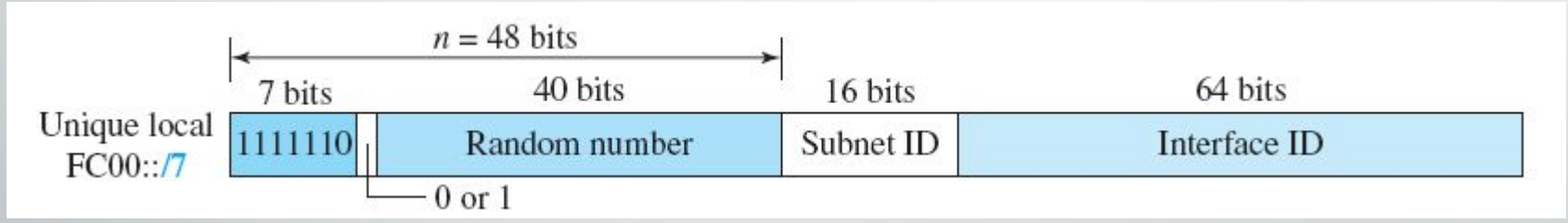
Special Addresses



- **Unspecified Address:**
  - ::/128
  - In a host, it refers to the host itself, and is used when a device does not know its own address
  - For addressing purposes within a software. .
- **Loopback Address**
  - ::1/128
  - loopback (same as 127.0.0.1 in many IPv4 implementations)
  - In IPv6 there is just one address, not a whole block, for this function.

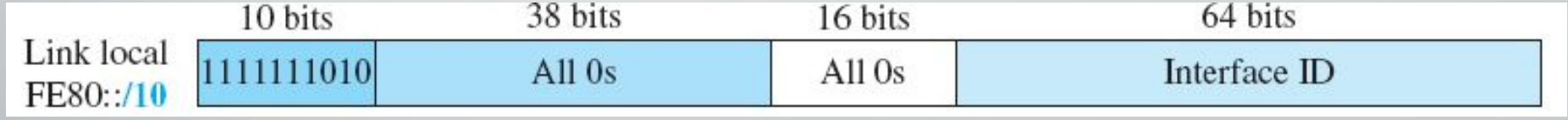
Unique Local Unicast Address

- **FCoo::/7**
- Globally unique,
- But it should be used in local communication.



Link Local Unicast Address

- **FE80::/10**
  - These addresses refer only to a particular physical network.
  - Routers do not forward datagrams using link-local addresses.
  - They are only for local communication on a particular physical network segment.
    - Automatic address configuration.
    - Neighbor discovery.

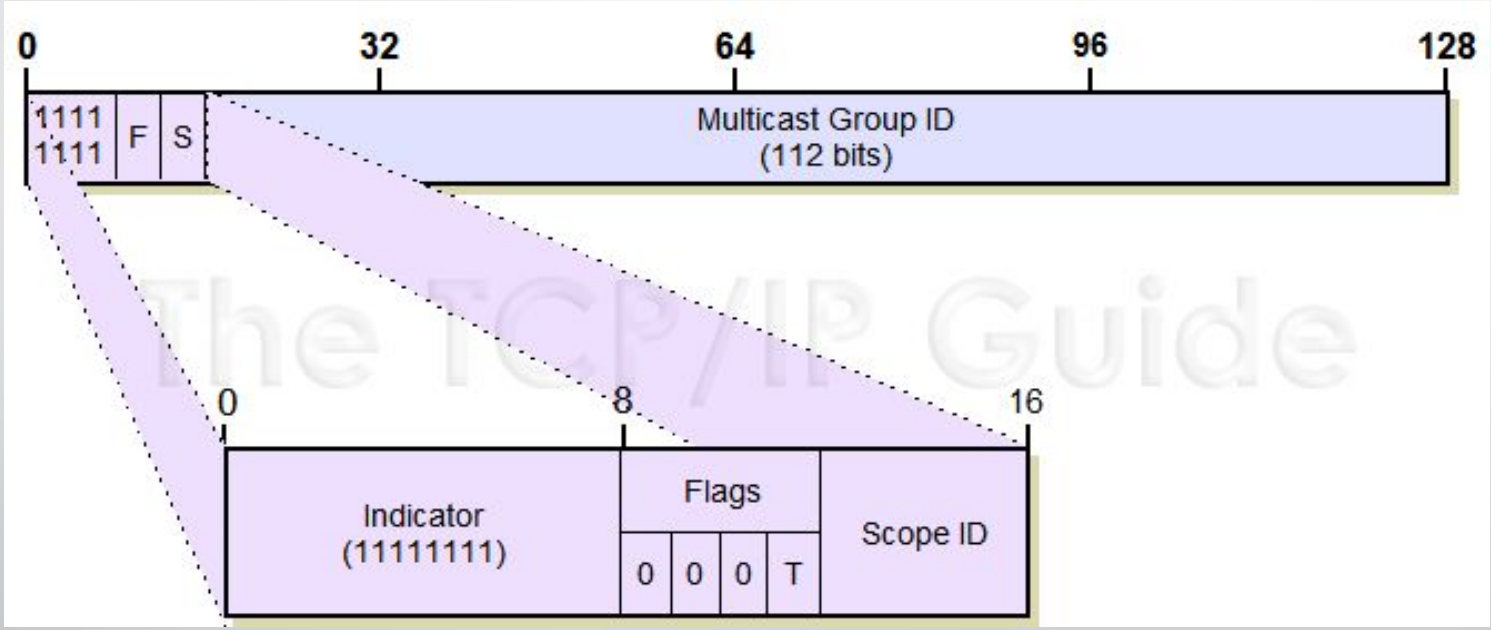


Scope of IPv6 Unicast Addresses



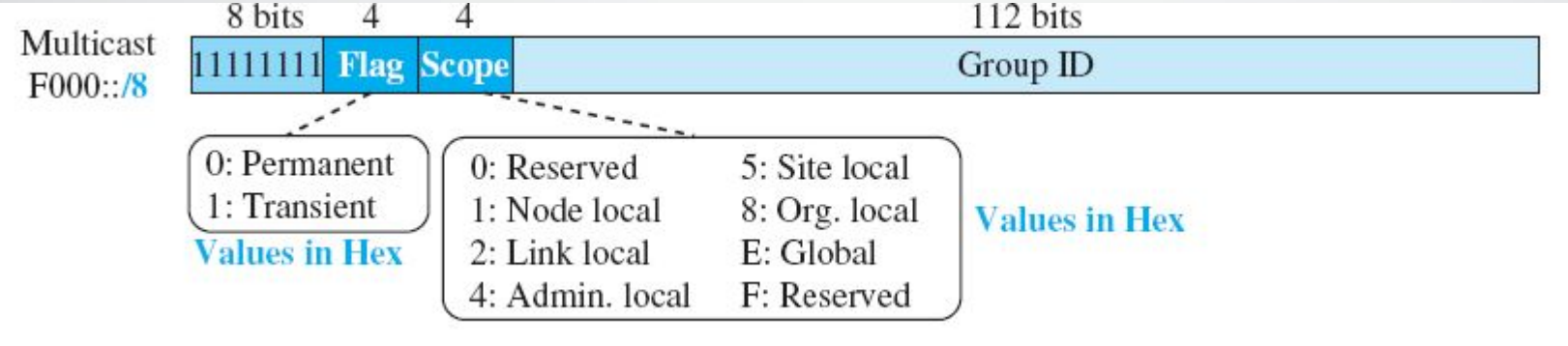
Multicast Addresses

- Consisting of all addresses that begin with "1111 1111" i.e "FF"



Multicast Addresses

- Multicast addresses are used to send data to a number of devices on an internetwork simultaneously.
- Each multicast address can be specified for a variety of different *scopes*
  - allowing a transmission to be targeted to either a wide or narrow audience of recipient devices.



Anycast Addresses

- To provide flexibility in situations where we need a service that is provided by a number of different servers or routers but don't really care which one provides it.
- In routing, anycast allows datagrams to be sent to whichever router in a group of equivalent routers is closest



Anycast Addresses

- There is no special anycast addressing scheme: anycast addresses are the same as unicast addresses.
- An anycast address is created "automatically" when a unicast address is assigned to more than one interface.

Transition from IPv4 to IPv6

IPv4 to IPv6 Transition

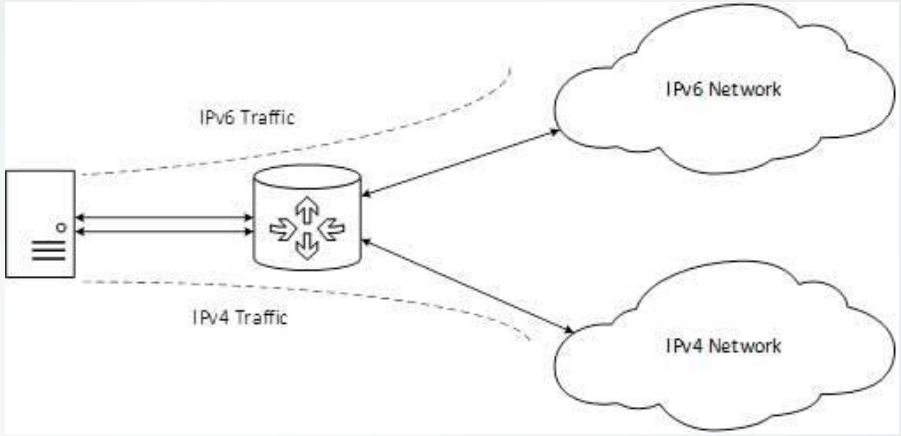
Strategies and mechanisms:

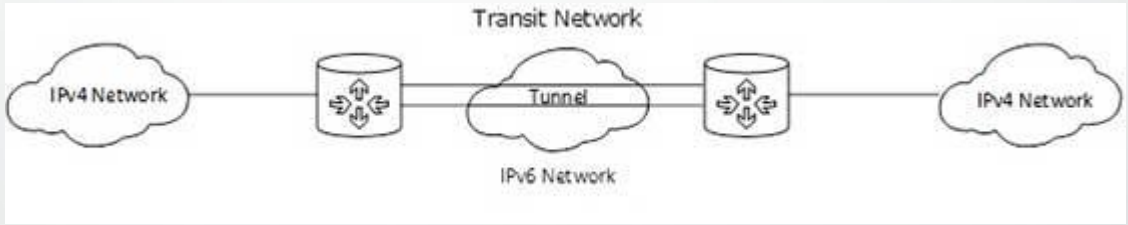
- IPv4 to IPv6 transition is gradual
- IPv6 devices need to communicate to IPv4
- IPv6 needs to communicate over IPv4 links

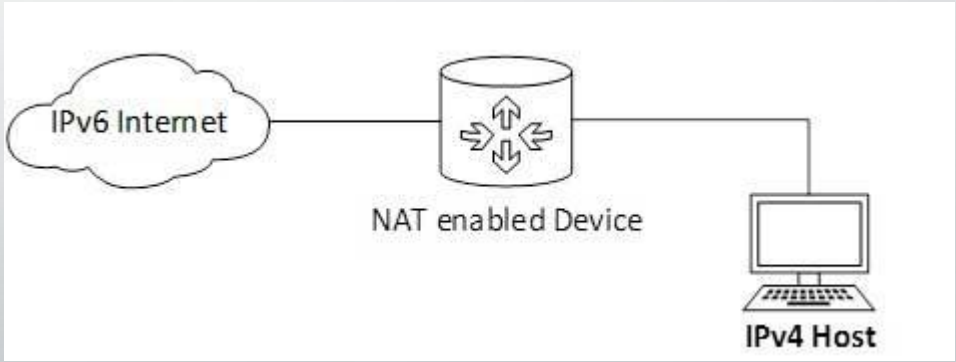
Transition Techniques

Three categories:

- **Dual-stack techniques**
- **Tunneling Techniques**
- **Translation techniques**

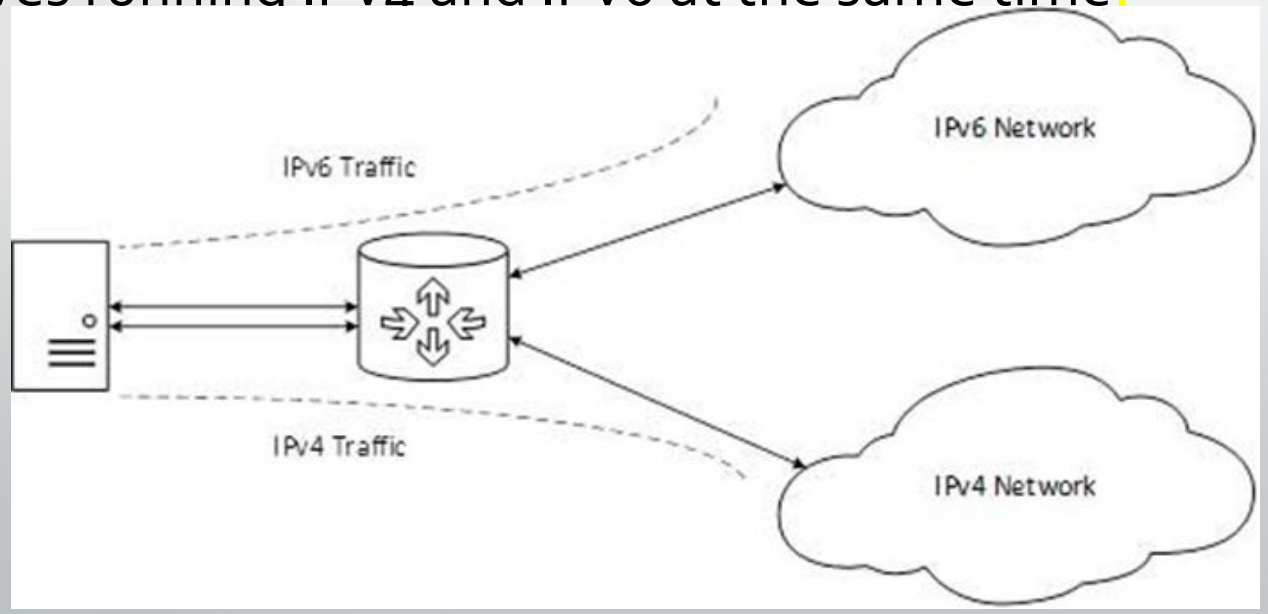




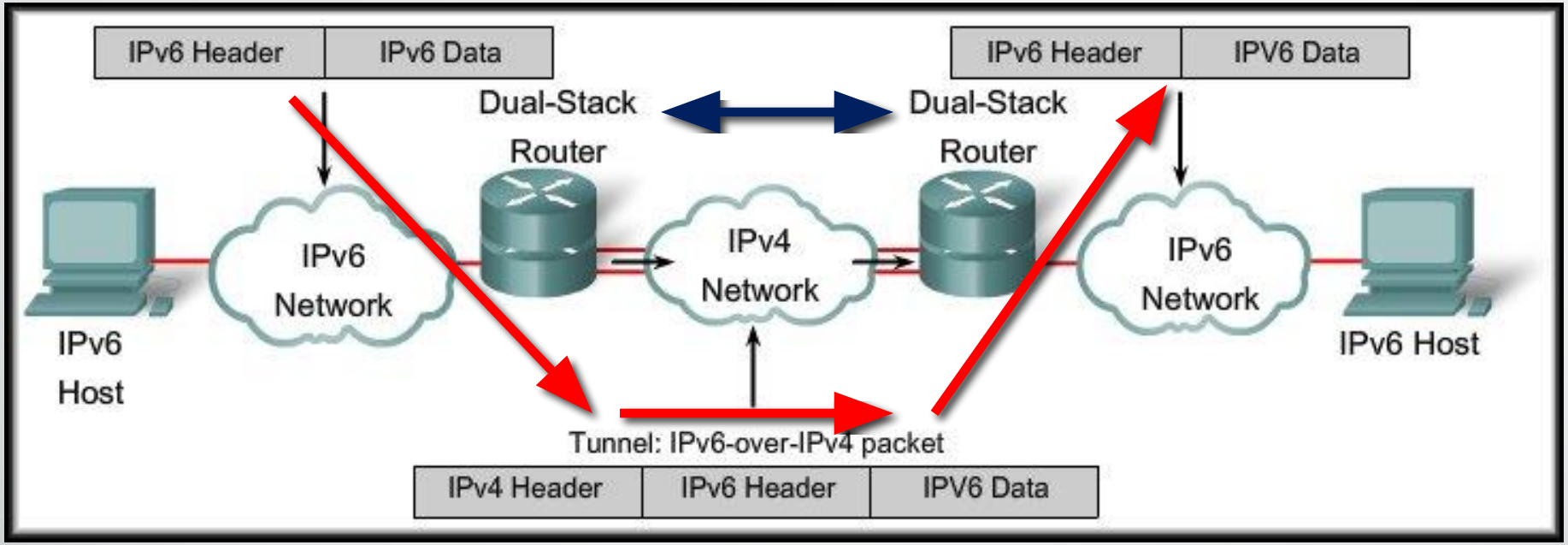


Dual Stack

- Method in which a node has implementation and connectivity to both an IPv4 and IPv6 network.
- The recommended option.
- Involves running IPv4 and IPv6 at the same time.

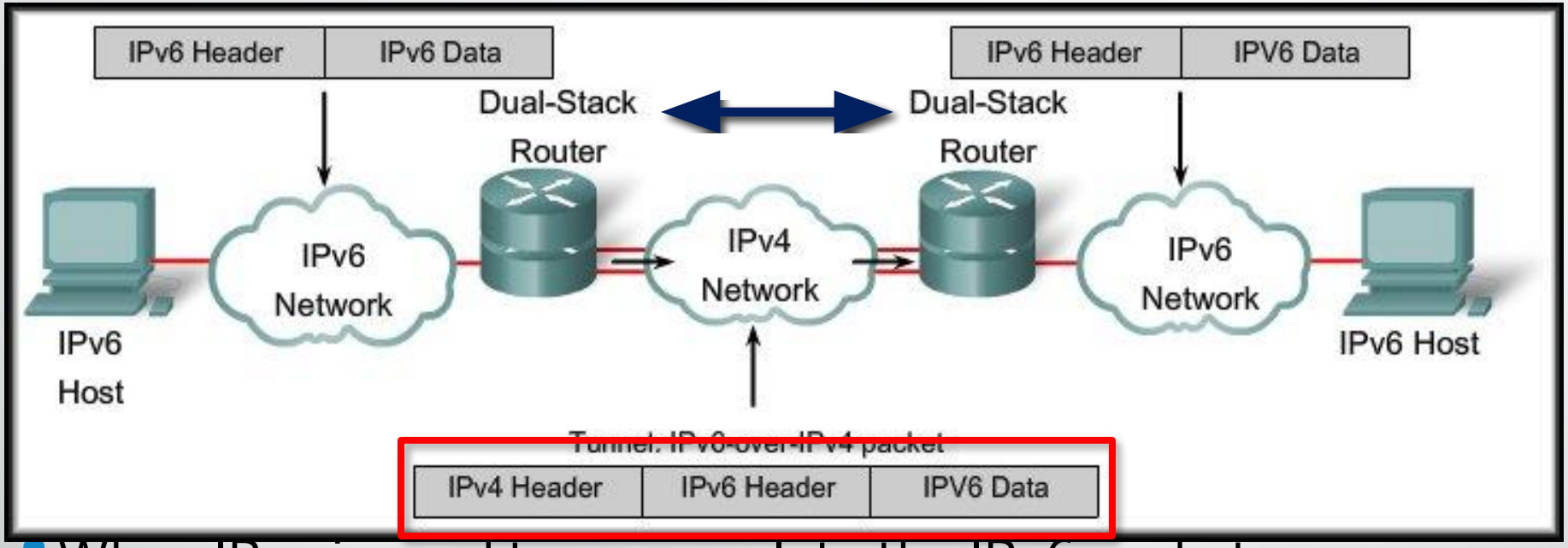


IPv6 Tunneling



- Tunneling is an integration method where an IPv6 packet is encapsulated within another protocol.
- Tunneling encapsulates the IPv6 packet in the IPv4 packet.

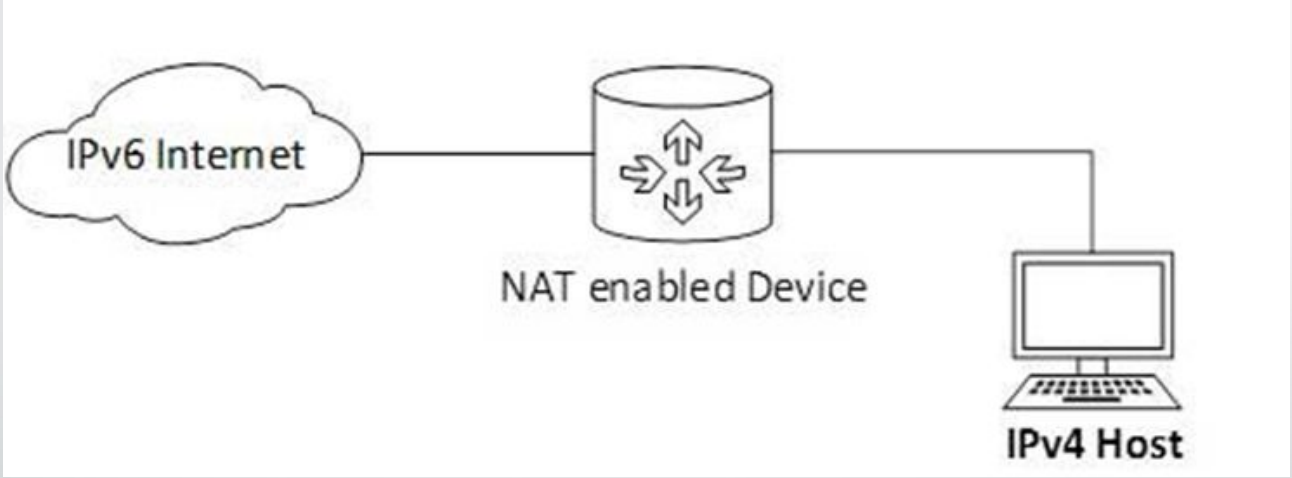
IPv6 Tunneling



- When IPv4 is used to encapsulate the IPv6 packet:
  - Protocol type of 41.
  - 20-byte IPv4 header with no options.
  - IPv6 header and payload.
  - Requires dual stacked routers.

NAT Protocol Translation

- Important method of transition to IPv6 by means of a NAT-PT (Network Address Translation – Protocol Translation) enabled device.



When the IPv4 host sends a request packet to the IPv6 server, the NAT-PT device/router strips down the IPv4 packet, removes IPv4 header, and adds IPv6 header and passes it through the Internet. When a response from the IPv6 server comes for the IPv4 host, the router does vice versa.

THE END